AI Investment Challenge

Stage 1 Concept Submission | 2025-2026

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1. Problem Definition

In the current investment landscape, specifically in Equity Research, Asset Management, Private Equity and Long/Short Fundamental Hedge Funds, analysts build DCFs and operating models using traditional financial accounting standards to forecast long-term value. Revenue growth, operating margins, capital expenditure and working capital changes drive the model. The intangibles, including how a company manages its environmental risks, governance structures or social obligations, are acknowledged but rarely quantified, leaving a significant dimension of long-term financial risk outside the model entirely.

This is not a new observation. Edmans (2023, p.3) argues ESG factors are neither more nor less important than other intangible value drivers such as management quality or corporate culture, and that any serious investor should treat them with the same rigour. The difficulty is practical. ESG data is largely voluntary, self-reported and unstandardised across reporting frameworks. A junior analyst trying to assess a company's water management risk or supply chain exposure has no reliable, structured starting point. Cross-referencing sustainability reports, CDP filings and regulatory disclosures manually takes days, and even then the accuracy and consistency of the data is unclear.

The problem is compounded structurally. Many asset managers approach ESG through a single lens, typically whatever their fund mandate highlights, without considering which factors are actually material to that specific business. A data privacy concern is financially significant for a technology company in a way it simply is not for an oil major. Generic ESG scoring tools apply the same metrics to every company regardless of industry, producing comparisons that miss the risks that matter. PwC (2022) reports 79% of investors consider ESG management important to their decision-making, yet Berg et al. (2022) show correlations between major ESG rating providers are as low as 0.38 for the same company.

Two specific gaps remain unaddressed by existing tools. First, there is no systematic cross-check between what companies voluntarily publish in sustainability reports and what they are legally required to disclose to regulators under separate obligations. These two documents are produced by different teams for different audiences and frequently diverge in ways that carry direct financial implications. Second, even when analysts identify ESG risks, no tool maps those risks to the specific lines of the financial model they have already built. ESG Signal addresses both gaps directly, scoped to FTSE 100 and S&P 500 companies where the depth of publicly available regulatory data makes rigorous independent verification possible.

2. Solution Overview

ESG Signal takes two inputs: a FTSE 100 or S&P 500 company ticker and an Excel DCF model uploaded locally. It produces an ESG Commitment Credibility Briefing identifying which ESG risks are material for that company's specific industry, how credible the company's management of those risks is based on five independent evidence streams, and which line items in the analyst's model are exposed to each risk alongside scenario ranges sourced from real regulatory and financial precedents. All DCF processing runs locally and the model never leaves the analyst's machine for security. Corporate sustainability reports remain voluntary for most companies, meaning data availability varies across the index. This is partially mitigated by cross-referencing against mandatory regulatory submissions, and will improve as CSRD and FCA SDR requirements progressively mandate disclosure from 2025 onwards.

Agent 1: Data Gathering

Collects publicly available data across five source types: annual reports and risk factor sections from the SEC EDGAR free API for S&P 500 companies and Companies House for FTSE 100; sustainability reports in PDF format processed by LayoutParser; standardised climate disclosures from the CDP Open Data Portal; news coverage from the GDELT Project for external signals including regulatory fines and environmental incidents; and capital expenditure and operating expense breakdowns from financial statements for the Words vs Money cross-check.

Agent 2: ESG Relevance Filter

Applies the SASB Materiality Map to identify which ESG factors are financially material for the specific company's industry across 77 sectors, distinguishing factors that plausibly affect revenues, costs or assets from those that are not relevant to the business model. Industry classification is derived automatically from EDGAR SIC codes for S&P 500 companies and SIC codes from Companies House for FTSE 100. This filters analysis to the factors that actually matter for that business, which is the step that makes the briefing specific rather than generic.

Agent 3: Multi-Source Verification and Credibility Scoring

Each material ESG factor is cross-referenced across five independent evidence streams producing a per-factor Commitment Credibility Score.

The first stream is disclosure quality, flagging each factor as quantified, vague or undisclosed with year-over-year drift detection identifying methodology changes between reporting periods. The second is the Words vs Money check, comparing capital expenditure and operating expense lines from financial statements against stated ESG commitments in Python. The third is the Talent Signal, analysing job postings scraped from publicly accessible sources filtered by SASB-mapped ESG keywords over a rolling 24-month window, with ghost listing detection via posting lifecycle tracking at 30-day intervals, seniority distribution analysis and LinkedIn public headcount cross-referencing. The fourth is Mandatory Disclosure Cross-Referencing, the core independent verification layer. For S&P 500 companies this draws on the EPA Greenhouse Gas Reporting Program for facility-level emissions under federal legal obligation, the NRC incident database for legally mandated spill and release reporting, the EPA ECHO enforcement database for violation and penalty history, SEC 10-K risk factor disclosures for legally mandated environmental liability statements, and state environmental agency databases covering TCEQ, CARB and PADEP for the major industrial jurisdictions. For FTSE 100 companies this draws on the UK Environment Agency Pollution Inventory for UK facility operations, EU ETS independently verified submissions covering European industrial facilities, and where FTSE 100 companies have US operations, the full EPA stack above. Because these mandatory disclosures are produced under separate legal obligations from sustainability reports, divergence between them and voluntary ESG claims is a specific, sourced signal rather than an opinion. The fifth stream is supply chain exposure, identifying top named suppliers from regulatory filings and flagging their SASB material factor profile for Scope 3 risk assessment.

DCF Line Item Mapping

Where an analyst uploads a DCF, openpyxl parses the line items locally and the tool maps each identified ESG risk to the specific lines most likely to be affected using SASB financial impact categories. Scenario ranges are generated from real precedents: EPA enforcement amounts from the ECHO database, EU ETS penalty schedules, UK EA enforcement records, LSEG public financial analysis and statutory filings, and historical impairments from comparable transitions in public financial statements. The analyst sees exactly where in their model each risk sits and what comparable situations have produced.

Output

A Streamlit web interface with colour-coded confidence indicators. Green for corroborated across multiple independent sources. Amber for inconsistently supported. Red for undisclosed or contradicted by mandatory regulatory data. Every signal cites its primary source. Exportable as PDF and structured JSON.

Figure 1 - Analysis Pipeline: how a ticker and DCF become a Credibility Briefing

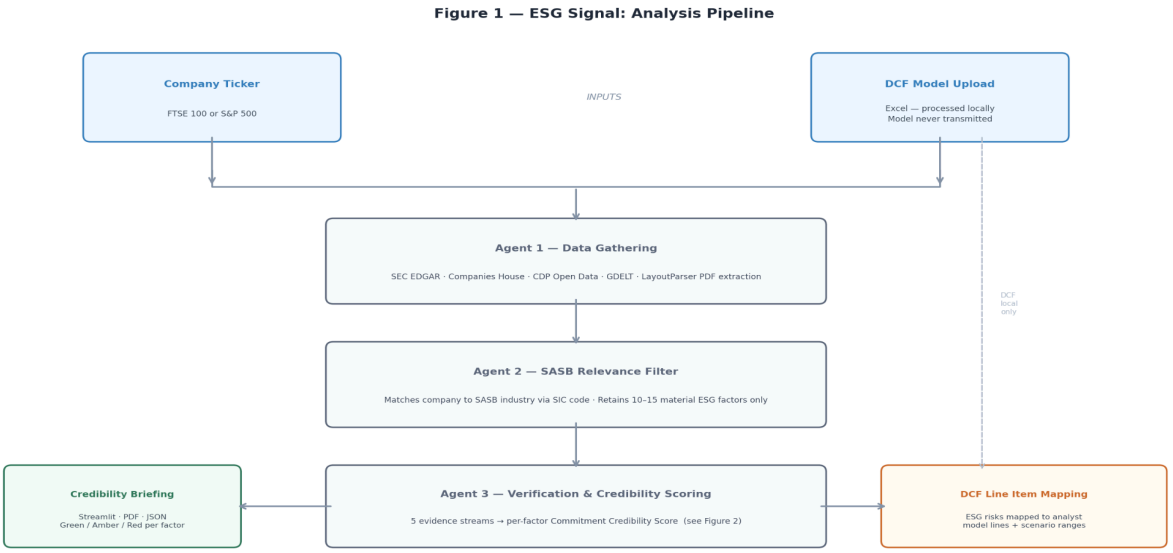
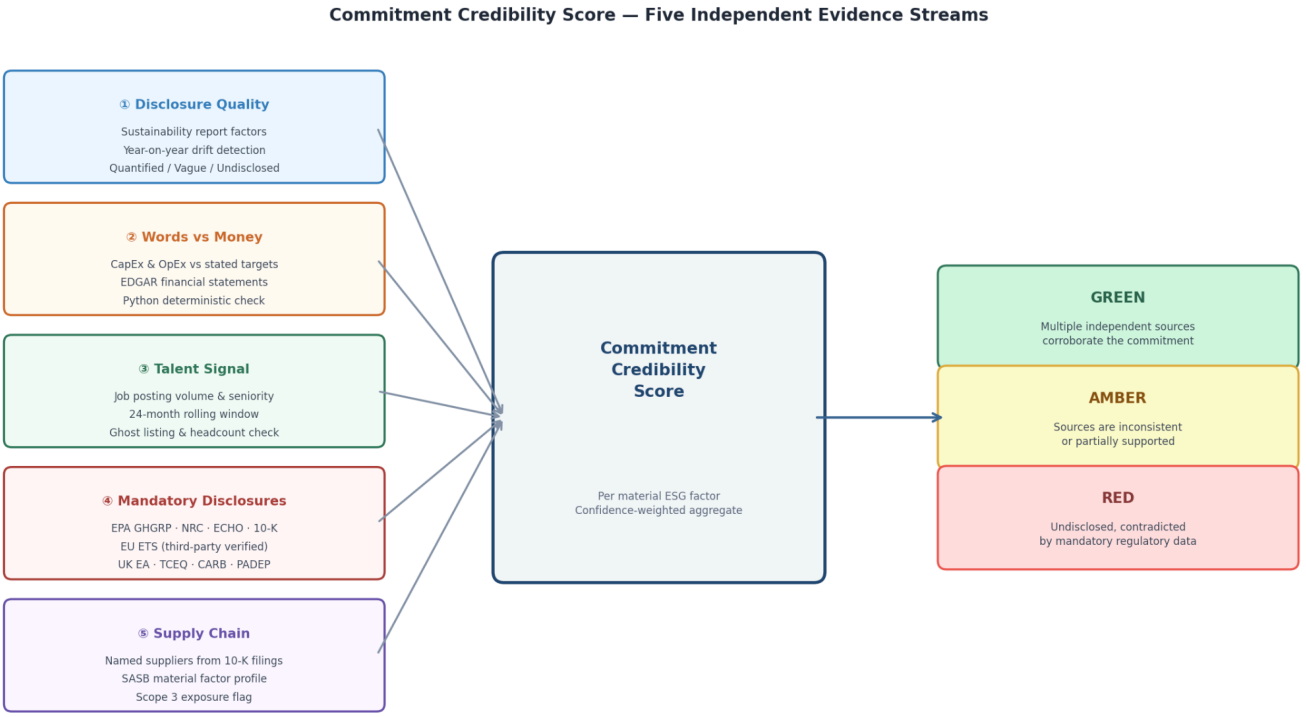


Figure 2 - Commitment Credibility Score: five independent evidence streams



3. Planned Implementation

Built in Python with Apache Airflow orchestrating the full pipeline, providing sequential task management, API retry handling and a complete audit log for reproducibility under Rule 4.4. PDF extraction uses LayoutParser for multi-column sustainability report layouts that are relayed into an OCR tool. Google Gemini handles long document processing on its free tier. Anthropic's Claude Opus 4.5 model handles structured briefing synthesis and DCF line item mapping. All AI calls are logged with model, version and token cost per Rule 4.5. Estimated per-run cost is under five dollars. All numerical calculations, drift comparisons, financial cross-checks and confidence aggregations run in Python via a deterministic validation layer, not the LLM, preventing hallucinated figures from appearing in output.

EPA GHGRP and ECHO data are downloaded as structured CSV via free EPA APIs. UK EA Pollution Inventory and EU ETS data are downloaded as annual structured files from their respective public portals. Companies House filings are accessed via the free Companies House API. LSEG public financial data and statutory filings supplement DCF scenario generation. Sentinel-2 NDVI queries for operational footprint verification run via the geemap Python library against Google Earth Engine with computation on Google's servers. DCF parsing uses openpyxl locally. Talent Signal scraping uses the Indeed API free tier and BeautifulSoup for publicly accessible LinkedIn data with 30-day re-query intervals for lifecycle tracking.

The pilot covers four SASB industries across both indices: integrated oil and gas, commercial banking, software and IT services, and food and beverage, chosen for data depth and to demonstrate the tool across industries with genuinely different material ESG profiles. Physical verification draws on all available mandatory sources for each company's jurisdictional footprint. Where operations fall outside covered regulatory frameworks, confidence scores are explicitly reduced with plain-English disclosure. All code is MIT licensed and publicly hosted. Development follows agile methodology with weekly sprints and peer code review before any merge to main.

4. Impact Assessment

The primary user is a junior analyst, typically zero to three years into their career, responsible for building and maintaining financial models across a portfolio of FTSE 100 or S&P 500 companies. This analyst is most likely to be asked to incorporate ESG factors into a valuation and least likely to have the time or resources to do so rigorously. Cross-referencing sustainability reports, CDP filings and regulatory disclosures for a single company manually takes between four and eight hours. ESG Signal reduces this to under ten minutes while extending the analysis to independent mandatory disclosure sources no manual review could cost-effectively cover.

The quality improvement matters more than the time saving. A manual review surfaces what the company wants analysts to see. ESG Signal surfaces what is not prominently presented: material factors not disclosed against, capital expenditure inconsistent with stated commitments, hiring patterns that do not support announced programmes, and most significantly, divergences between voluntary sustainability reports and legally mandated regulatory submissions. No existing tool performs this cross-check systematically for both FTSE 100 and S&P 500 companies. The DCF mapping layer closes the final gap. Rather than leaving the analyst to figure out what an ESG signal means for their model, ESG Signal maps each risk to the specific lines in their own DCF with scenario ranges sourced from real enforcement records and historical financial comparables.

For Portfolio Managers running Long/Short strategies, the mandatory disclosure cross-reference provides early warning of credibility gaps before they become public events. A FTSE 100 energy company whose EU ETS verified submissions diverge from its sustainability report trajectory is carrying a specific, documented risk that most investors are not systematically checking. At the industry level, ESG Signal directly supports compliance with FCA Sustainability Disclosure Requirements and EU CSRD, both of which demand substantive, auditable ESG analysis from UK asset managers rather than reliance on opaque third-party scores.

5. Ethical Considerations

All external data is publicly available and fully documented with source URLs. When an analyst uploads a DCF model, that file is processed entirely locally and never transmitted to any server, protecting proprietary investment model confidentiality and ensuring the tool is viable within professional compliance constraints. The principal ethical risk is hallucination. This is addressed through the deterministic validation layer: all numerical outputs are calculated in Python from structured data, the LLM produces only qualitative narrative and line item mapping, and every claim cites its primary source. DCF scenario ranges trace exclusively to cited regulatory records and public financial statements, not LLM generation.

Data bias is mitigated by cross-referencing voluntary disclosures against multiple independent mandatory regulatory sources. Where sources disagree the tool flags the conflict and lowers the confidence score rather than averaging it away. The mandatory disclosure cross-reference layer is explicitly scoped to jurisdictions where independent regulatory data exists, with confidence scores on emissions factors transparently reduced for operations outside those jurisdictions and a plain-English explanation provided. The Talent Signal addresses ghost listings through lifecycle tracking, seniority analysis and headcount cross-referencing before assigning weight to hiring data. The tool is designed as decision support only. It does not make investment recommendations and includes a clear disclaimer in every output. Analysts retain full responsibility for any investment decisions made using the briefing.

References

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